

# MTH 490: Supremum and Infimum

## Problem 23

Theresa Wunderlich

### 1.4 Supremum, Infimum, and Completeness

**Definition 1.8.** Let  $A \subset \mathbb{R}$  and  $a \in \mathbb{R}$ .

- (i)  $a$  is an **upper bound** of  $A$  if  $x \leq a$  for all  $x \in A$ . If  $a$  is an upper bound of  $A$  and  $a \in A$ , we say  $a$  is a **maximum** of  $A$ .
- (ii)  $a$  is a **lower bound** of  $A$  if  $a \leq x$  for all  $x \in A$ . If  $a$  is a lower bound of  $A$  and  $a \in A$ , we say  $a$  is a **minimum** of  $A$ .
- (iii) If a set  $A$  has an upper bound we say it is **bounded above**. If  $A$  has a lower bound we say it is **bounded below**. If  $A$  is bounded both above and below, we say it is **bounded**.

**Definition 1.9.** If  $u$  is an upper bound for the set  $A$  such that  $u \leq a$  for all upper bounds  $a$  of  $A$ , then  $u$  is called a **supremum** or **least upper bound** of  $A$ .

**Definition 1.10.** If  $l$  is a lower bound for the set  $A$  and  $l \geq a$  for all lower bounds  $a$  of  $A$ , then  $l$  is called an **infimum** or **greatest lower bound** of  $A$ .

**23.** Find the min, max, infimum and supremum (if they exist) of the following subsets of  $\mathbb{R}$ :

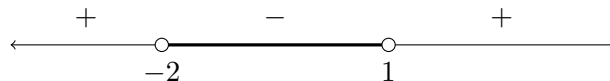
- (a)  $\{x : x^2 + x - 2 < 0\}$
- (b)  $\{x \in \mathbb{Q} : x^2 \leq 2\}$
- (c)  $(0, \infty)$
- (d)  $\left\{\frac{n}{n+m} : m, n \in \mathbb{N}\right\}$
- (e)  $\left\{\frac{n}{m} : m, n \in \mathbb{N} \text{ with } m + n \leq 10\right\}$
- (f)  $\left\{\frac{1}{n} + (-1)^n : n \in \mathbb{N}\right\}$

For each part I sketch or test the set, then find the bounds. The sup is the smallest upper bound and the inf is the largest lower bound. A max or min only exists if the sup or inf is actually in the set. Here  $\mathbb{N} = \{1, 2, 3, \dots\}$ .

*Solution to (a).* Set  $x^2 + x - 2 = 0$  to find the endpoints:

$$(x - 1)(x + 2) = 0 \implies x = 1 \text{ or } x = -2.$$

Checking the sign of  $(x - 1)(x + 2)$  on each side of the roots:



The product is only negative between the roots. At  $x = -2$  and  $x = 1$  it equals 0, so the endpoints are not included. The set is  $(-2, 1)$ .

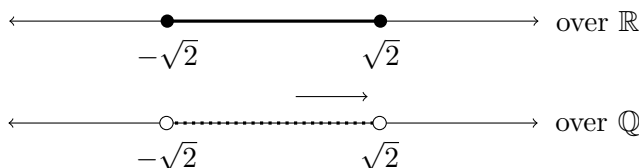
Every number  $\geq 1$  is an upper bound. A number less than 1 is not, since there are points of the set between it and 1. So the smallest upper bound is 1. In the same way, every number  $\leq -2$  is a lower bound, and the largest one is  $-2$ .

$$\sup = 1, \quad \inf = -2, \quad \text{no max}, \quad \text{no min}$$

There is no max or min because 1 and  $-2$  are not in the set.  $\square$

*Solution to (b).* Setting  $x^2 = 2$  gives  $x = \pm\sqrt{2}$ . So the set is the rational numbers between  $-\sqrt{2}$  and  $\sqrt{2}$ .

Comparing to the same set over  $\mathbb{R}$ :

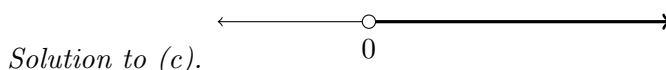


Over  $\mathbb{R}$  the endpoints are included, so  $\sqrt{2}$  would be the max. Over  $\mathbb{Q}$  they are not included, because  $\sqrt{2}$  is irrational.

If  $x > \sqrt{2}$  then  $x^2 > 2$  (Problem 16), so  $\sqrt{2}$  is an upper bound. Nothing smaller works, because the rationals  $1.4, 1.41, 1.414, \dots$  are all in the set and keep getting closer to  $\sqrt{2}$ . So  $\sup = \sqrt{2}$ . The set is symmetric about 0, so  $\inf = -\sqrt{2}$ .

$$\sup = \sqrt{2}, \quad \inf = -\sqrt{2}, \quad \text{no max}, \quad \text{no min}$$

There is no max or min because  $\pm\sqrt{2}$  are not rational.  $\square$



*Solution to (c).*

Every number  $\leq 0$  is a lower bound. A positive number  $l$  is not, since  $\frac{l}{2}$  is in the set and smaller. So  $\inf = 0$ . Since 0 is not in the set, there is no min.

The set keeps going to the right forever, so there is no upper bound.

$$\inf = 0, \quad \text{no min}, \quad \text{no sup}, \quad \text{no max}$$

$\square$

*Solution to (d).* Compare  $\frac{n}{n+m}$  to  $\frac{1}{2}$ :

$$\frac{n}{n+m} < \frac{1}{2} \iff 2n < n+m \iff n < m.$$

So there are three cases:

$$\bullet \quad n = m: \quad \frac{n}{2n} = \frac{1}{2}$$

- $n < m$ : less than  $\frac{1}{2}$
- $n > m$ : greater than  $\frac{1}{2}$

Testing values:

| $n, m$          | $\frac{n}{n+m}$  | vs. $\frac{1}{2}$ |
|-----------------|------------------|-------------------|
| $n = m = 1$     | $\frac{1}{2}$    | equal             |
| $n = 1, m = 3$  | $\frac{1}{4}$    | less              |
| $n = 4, m = 1$  | $\frac{4}{5}$    | greater           |
| $n = 1, m = 99$ | $\frac{1}{100}$  | less              |
| $n = 99, m = 1$ | $\frac{99}{100}$ | greater           |

When  $m$  is big the values get close to 0, and when  $n$  is big they get close to 1.

Since  $0 < n < n + m$ , every value is between 0 and 1. With  $n = 1$ , the values  $\frac{1}{1+m}$  get as close to 0 as we want (Problem 28), so  $\inf = 0$ . With  $m = 1$ , the values  $\frac{n}{n+1}$  get as close to 1 as we want, so  $\sup = 1$ .

$$\sup = 1, \quad \inf = 0, \quad \text{no max}, \quad \text{no min}$$

There is no max or min because no value actually equals 0 or 1. □

*Solution to (e).* Since  $m + n \leq 10$ , there are only finitely many pairs, so the set has a max and a min. Testing the extremes:

| $n, m$         | $\frac{n}{m}$ |
|----------------|---------------|
| $n = 9, m = 1$ | 9             |
| $n = 5, m = 5$ | 1             |
| $n = 1, m = 9$ | $\frac{1}{9}$ |

The fraction is biggest with the largest top and smallest bottom:  $n = 9, m = 1$  gives 9. Since  $n \leq 9$  and  $m \geq 1$ , we always have  $\frac{n}{m} \leq 9$ .

It is smallest with the smallest top and largest bottom:  $n = 1, m = 9$  gives  $\frac{1}{9}$ . Since  $n \geq 1$  and  $m \leq 9$ , we always have  $\frac{n}{m} \geq \frac{1}{9}$ .

$$\max = \sup = 9, \quad \min = \inf = \frac{1}{9}$$

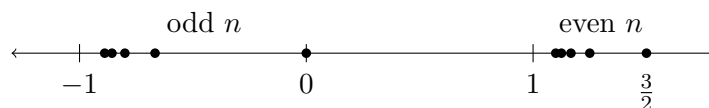
□

*Solution to (f).*  $(-1)^n$  depends on whether  $n$  is even or odd:

- $n$  even: the term is  $1 + \frac{1}{n}$
- $n$  odd: the term is  $-1 + \frac{1}{n}$

Testing values:

| $n$ | term              | value          |
|-----|-------------------|----------------|
| 1   | $1 - 1$           | 0              |
| 2   | $\frac{1}{2} + 1$ | $\frac{3}{2}$  |
| 3   | $\frac{1}{3} - 1$ | $-\frac{2}{3}$ |
| 4   | $\frac{1}{4} + 1$ | $\frac{5}{4}$  |
| 5   | $\frac{1}{5} - 1$ | $-\frac{4}{5}$ |
| 6   | $\frac{1}{6} + 1$ | $\frac{7}{6}$  |



The even terms go down toward 1, and the odd terms go down toward  $-1$ .

The biggest term is at  $n = 2$ :  $\frac{3}{2}$ . Every other even term is smaller, and every odd term is at most 0. So  $\max = \frac{3}{2}$ .

Every term is greater than  $-1$ , since  $\frac{1}{n} > 0$ . The odd terms get as close to  $-1$  as we want (Problem 28), so  $\inf = -1$ . No term equals  $-1$ , so there is no min.

$$\max = \sup = \frac{3}{2}, \quad \inf = -1, \quad \text{no min}$$

□